Chapter 2 Determinants

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Chapter 2 Determinants

Section 2.1 The Determinant of a Matrix

cofactor expansion

Definition (Minor and Cofactor) Let $A = (a_{ij})$ be an $n \times n$ matrix, and let M_{ij} denote the $(n-1) \times (n-1)$ matrix obtained from A by deleting the row and column containing a_{ij}. The determinant of M_{ij} is called the **minor** of a_{ij}.
We define the **cofactor** A of a_{ij} by

$$A_{ij} = (-1)^{i+j} \det(M_{ij}).$$

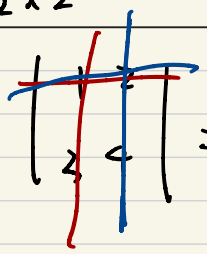
cofactor

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} \end{bmatrix}, \quad M_{ij} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} \end{bmatrix}$$

計算步驟有 $n!$

2×2

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - cb$$



$= 1 \times A_{11} + 2 \times A_{12} = 1 \cdot 4 + 2 \cdot (-3) = -2$

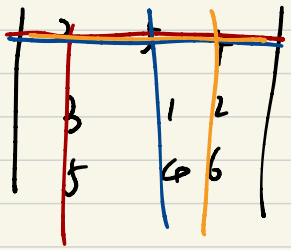
(1 = row, 1 = column)

delete the row & column.

$A_{11} = (-1)^{1+1} \det M_{11} = 1 \cdot \det[4] = 4$

$A_{12} = (-1)^{1+2} \det M_{12} = -1 \cdot \det[3] = -3$

3×3



$= 3 \times (2\times 2) - 5 \cdot \begin{vmatrix} 1 & 2 \\ 4 & 6 \end{vmatrix} + 4 \cdot \begin{vmatrix} 3 & 1 \\ 5 & 4 \end{vmatrix}$

$(-1)^{1+1}$; $(-1)^{1+2}$; $(-1)^{1+3}$

$= 2(6-8) - 5(18-10) + 4(12-5) = 16$

exercise :

$$\begin{vmatrix} 0 & 0 & 3 \\ 0 & 4 & 1 \\ 2 & 3 & 1 \end{vmatrix} = 0 - 0 + 3 \begin{vmatrix} 0 & 4 \\ 2 & 3 \end{vmatrix}$$

$= 3(0-8) = -24$

Definition (Determinant) The **determinant** of an $n \times n$ matrix A, denoted $\det(A)$ or $|A|$, is a scalar associated with the matrix A that is defined inductively as

$$\det(A) = \begin{cases} a_{11} & \text{if } n = 1 \\ a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1n}A_{1n} & \text{if } n > 1 \end{cases},$$

where

$$A_{1j} = (-1)^{1+j} \det(M_{1j}) \qquad j = 1, 2, \cdots, n$$

are cofactors associated with the entries in the first row of A.

Example Find $\det(A)$ and $\det(B)$ for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 5 & 4 \\ 3 & 1 & 2 \\ 5 & 4 & 6 \end{pmatrix}$.

Solution

$$\det(A) = 1 \times 4 - 2 \times 3 = -2$$

$$\begin{aligned} \det(B) &= b_{11} \det(M_{11}) - b_{12} \det(M_{12}) + b_{13} \det(M_{13}) \\ &= 2 \begin{vmatrix} 1 & 2 \\ 4 & 6 \end{vmatrix} - 5 \begin{vmatrix} 3 & 2 \\ 5 & 6 \end{vmatrix} + 4 \begin{vmatrix} 3 & 1 \\ 5 & 4 \end{vmatrix} \\ &= 2(6-8) - 5(18-10) + 4(12-5) \\ &= -16 \end{aligned}$$

Exercise*

Find the determinant of $C = \begin{pmatrix} 0 & 0 & 3 \\ 0 & 4 & 1 \\ 2 & 3 & 1 \end{pmatrix}$.

Answer: -24.

Remark

If A is a 2×2 matrix, then

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Theorem If A is an $n \times n$ matrix with $n \geq 2$, then $\det(A)$ can be expressed as a cofactor expansion **using any row or column** of A, i.e.

$$\begin{aligned} \det(A) &= a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in} \\ &= a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj} \end{aligned},$$

where $i = 1, 2, \cdots, n, \quad j = 1, 2, \cdots, n.$

$\rightarrow$ Reduce computational cost by using the row or column that contains the most zeros.

Example Compute $\det(C)$ for $C = \begin{pmatrix} 0 & 0 & 3 \\ 0 & 4 & 1 \\ 2 & 3 & 1 \end{pmatrix}$.

Example Let $B = \begin{pmatrix} 2 & 5 & 4 \\ 3 & 1 & 2 \\ 5 & 4 & 6 \end{pmatrix}$. B_{12} B_{22} B_{32}

To find $\det(B)$ by expanding along the second column, we have

$$\det(B) = -5\begin{vmatrix} 3 & 2 \\ 5 & 6 \end{vmatrix} + 1\begin{vmatrix} 2 & 4 \\ 5 & 6 \end{vmatrix} - 4\begin{vmatrix} 2 & 4 \\ 3 & 2 \end{vmatrix} = -5(18-10) + (12-20) - 4(4-12) = -16.$$

To find $\det(B)$ by expanding along the third row, we have

$$\det(B) = 5\begin{vmatrix} 5 & 4 \\ 1 & 2 \end{vmatrix} - 4\begin{vmatrix} 2 & 4 \\ 3 & 2 \end{vmatrix} + 6\begin{vmatrix} 2 & 5 \\ 3 & 1 \end{vmatrix} = 5(10-4) - 4(4-12) + 6(2-15) = -16.$$

B_{31} B_{32} B_{33}

Example

$$\begin{vmatrix} 2 & 3 & -1 & 3 & 3 \\ 0 & -1 & 5 & 2 & -1 \\ 0 & 0 & 3 & 9 & 2 \\ 0 & 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 0 & 5 \end{vmatrix} = 2(-1)^{1+1} \begin{vmatrix} -1 & 5 & 2 & -1 \\ 0 & 3 & 9 & 2 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{vmatrix}$$

$$= 2(-1)(-1)^{1+1} \begin{vmatrix} 3 & 9 & 2 \\ 0 & -1 & 3 \\ 0 & 0 & 5 \end{vmatrix}$$

$$= 2(-1)(3)(-1)^{1+1} \begin{vmatrix} -1 & 3 \\ 0 & 5 \end{vmatrix}$$

$$= 2(-1)(3)(-1)(-1)^{1+1} \begin{vmatrix} 5 \end{vmatrix}$$

$$= 2(-1)(3)(-1)(5) = 30$$

upper / lower triangular matrix

det (A)

Theorem If A is an $n \times n$ triangular (diagonal) matrix, then the determinant of A equals the product of the diagonal elements of A, i.e. $= (a_{11} \cdots a_{nn})$

$$\det \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = a_{11}a_{22}\cdots a_{nn}, \quad \det \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ 0 & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & b_{nn} \end{pmatrix} = \prod_{k=1}^{n} b_{kk}$$

Proof Exercise.

Theorem If A is a square matrix of order n, then $\det(A) = \det(A^T)$.

Proof The proof is by induction on n. Clearly, the result holds if $n = 1$, since 1×1 matrix must be a symmetric matrix.
Assume that the result holds for all $k \times k$ matrices.
Suppose that A is a $(k+1) \times (k+1)$ matrix. Expanding $\det(A)$ along the first row of A, we get

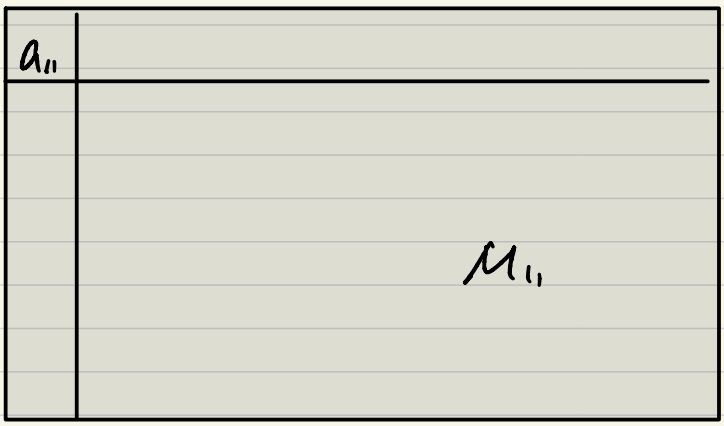
$$\det(A) = a_{11}\det(M_{11}) - a_{12}\det(M_{12}) + \cdots \pm a_{1,k+1}\det(M_{1,k+1}).$$

Since the M_{ij}'s are all $k \times k$ matrices, it follows from the induction hypothesis that

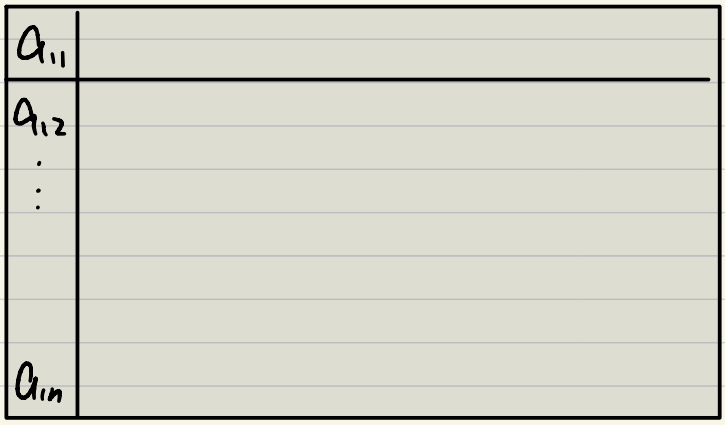
$$\det(A) = a_{11}\det(M_{11}^T) - a_{12}\det(M_{12}^T) + \cdots \pm a_{1,k+1}\det(M_{1,k+1}^T).$$

The right-hand side of the above equation is just the expansion by minors of $\det(A^T)$ using the first column of A^T. Therefore, $\det(A^T) = \det(A)$.

A:



A^T:



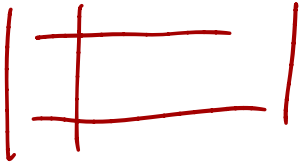
★前提很重要

Theorem Let A be an $n \times n$ matrix,

(i) If A has a row or column consisting entirely of zeros, then $\det(A) = 0$.

(ii) If A has two identical rows or two identical columns, then $\det(A) = 0$.

Proof Exercise.



the row switch => negative result

=> $\det(A) = -\det(A) \Rightarrow \det(A) = 0$

Exercise*

Compute $\det(A)$, $\det(B)$ and $\det(A+B)$ for $A = I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$.

$\det(A) = 1 \quad \det(B) = 1$

Remark

$\det(A+B) \neq \det(A) + \det(B)$ in general!!

how to prove?

$\det(A+B) = 0$

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = -1 \cdot \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} + 2 \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} - 3 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}$$

$$= -(45-48) + 2(36-42) - 3(32-35)$$

$$= 3 - 2 \times 6 + 3 \times 3$$

$$= 3 - 12 + 9 = 0 .$$

Chapter 2 Determinants

Section 2.2 Properties of Determinants

Type I Operation

When **switching** two rows (or columns) of $A_{n\times n}$: $R_i \leftrightarrow R_j$

$$\det\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = ??? \det\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

$$\det(EA) = ??? \det(A)$$

Type I Operation: Two rows of A are interchanged.

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Idea of Proof Use mathematical induction.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = -a_{21}\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{22}\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{23}\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\begin{vmatrix} a_{31} & a_{32} & a_{33} \\ a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \end{vmatrix} = -a_{21}\begin{vmatrix} a_{32} & a_{33} \\ a_{12} & a_{13} \end{vmatrix} + a_{22}\begin{vmatrix} a_{31} & a_{33} \\ a_{11} & a_{13} \end{vmatrix} - a_{23}\begin{vmatrix} a_{31} & a_{32} \\ a_{11} & a_{12} \end{vmatrix}$$

$$= -a_{21}\left(-\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}\right) + a_{22}\left(-\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}\right) - a_{23}\left(-\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}\right)$$

by induction assumption

Type I Operation: Two rows of A are interchanged.

Proposition Let A and E be $n \times n$ matrices. If E is a Type I elementary matrix, then $\det(EA) = \det(E)\det(A)$ where $\det(E) = -1$.

Idea of Proof Use mathematical induction.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = -a_{21}\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{22}\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{23}\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\begin{vmatrix} a_{31} & a_{32} & a_{33} \\ a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \end{vmatrix} = -a_{21}\begin{vmatrix} a_{32} & a_{33} \\ a_{12} & a_{13} \end{vmatrix} + a_{22}\begin{vmatrix} a_{31} & a_{33} \\ a_{11} & a_{13} \end{vmatrix} - a_{23}\begin{vmatrix} a_{31} & a_{32} \\ a_{11} & a_{12} \end{vmatrix}$$

$$= -a_{21}\left(-\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}\right) + a_{22}\left(-\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}\right) - a_{23}\left(-\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}\right)$$

by induction assumption

Type I Row/Col Op

When **switching** two rows (or columns) of $A_{n\times n}$: $R_i \leftrightarrow R_j$, $\det(E) =$ -1

$$\det\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = \text{-} \det\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = \text{-} \det(A).$$

$\det(EA) = \det(E)\det(A)$ where $\det(E) =$ -1.

Type II Operation

When **scaling** any row (or column) of $A_{n\times n}$ by a nonzero α: $\alpha R_i \to R_i$

$$\det\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ \alpha a_{i1} & \alpha a_{i2} & \cdots & \alpha a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = ??? \det\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

$$\det(EA) = ??? \det(A)$$

Type II Operation: A row of A is multiplied by a nonzero constant α.

Type II Operation: A row of A is multiplied by a nonzero constant α.

Proposition Let A be an $n \times n$ matrix. Let E denote the elementary matrix of Type II formed from I_n by multiplying the ith row by the nonzero constant α, then $\det(EA) = \det(E)\det(A)$ where $\det(E) = \alpha$.

Type II Operation: A row of A is multiplied by a nonzero constant α.

Proposition Let A be an $n \times n$ matrix. Let E denote the elementary matrix of Type II formed from I_n by multiplying the ith row by the nonzero constant α, then $\det(EA) = \det(E)\det(A)$ where $\det(E) = \alpha$.

Proof Expanding $\det(EA)$ by cofactors along the ith row given

$$\det(EA) = \alpha a_{i1}A_{i1} + \alpha a_{i2}A_{i2} + \cdots + \alpha a_{in}A_{in} = \alpha \det(A).$$

In particular, when $A = I$,

$$\det(E) = \det(EI) = \alpha \det(I) = \alpha$$

and hence,

$$\det(EA) = \alpha \det(A) = \det(E)\det(A).$$

Type II Row/Col Op

When **scaling** any row (or column) of $A_{n\times n}$ by a nonzero α: $\alpha R_i \to R_i$, $\det(E) = \alpha$

$$\det\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ \alpha a_{i1} & \alpha a_{i2} & \cdots & \alpha a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = \alpha \det\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

$$= \alpha \det(A)$$

$\det(EA) = \det(E)\det(A)$ where $\det(E) = \alpha$.

Type III Operation

When **replacing** any row (or col.) of A by the sum of a scalar multiple of another row(or col.) and itself: $\beta R_i + R_j \to R_j$

$$\det \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \cdots & \cdots & & \cdots \\ a_{j1} + \beta a_{i1} & a_{j2} + \beta a_{i2} & \cdots & a_{jn} + \beta a_{in} \\ \cdots & \cdots & & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

$$= ??? \det \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \cdots & \cdots & & \cdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \cdots & \cdots & & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

$$\det(EA) = ??? \det(A)$$

Type III Operation: A multiple of one row is added to another row.

Proposition Let A and E be $n \times n$ matrices. If E is a Type III elementary matrix, then $\det(EA) = \det(E)\det(A)$ where $\det(E) = 1$.

unchanged

Type III Operation: A multiple of one row is added to another row.

Proposition Let A and E be $n \times n$ matrices. If E is a Type III elementary matrix, then $\det(EA) = \det(E)\det(A)$ where $\det(E) = 1$.

Proof Let E be the elementary matrix of type III formed from I by adding c times the ith row to the jth row. Since E is triangular and its diagonal elements are all 1, it follows that $\det(E) = 1$. If $\det(EA)$ is expanded by cofactors along the jth row,

$$\begin{aligned}
\det(EA) &= (a_{j1} + ca_{i1})A_{j1} + (a_{j2} + ca_{i2})A_{j2} + \cdots + (a_{jn} + ca_{in})A_{jn} \\
&= (a_{j1}A_{j1} + \cdots + a_{jn}A_{jn}) + c(a_{i1}A_{j1} + \cdots + a_{in}A_{jn}) \\
&= \det(A) + c(0) \qquad \text{by the lemma on the next slide} \\
&= \det(A)
\end{aligned}$$

Thus,

$$\det(EA) = \det(A) = \det(E)\det(A).$$

still lower triangular form

Lemma Let A be an $n \times n$ matrix. If A_{jk} denotes the cofactor of a_{jk} for $k = 1, \cdots, n$, then

$$a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn} = \begin{cases} \det(A), & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases} \qquad (\sharp)$$

Lemma Let A be an $n \times n$ matrix. If A_{jk} denotes the cofactor of a_{jk} for $k = 1, \cdots, n$, then

$$a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn} = \begin{cases} \det(A), & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases} \qquad (\sharp)$$

Proof If $i = j$, $(\sharp)$ is just the cofactor expansion of $\det(A)$ along the ith row of A.
When $i \neq j$, let A^* be the matrix obtained by replacing the jth row of A by the ith row of A. Since two rows of A^* are the same, its determinant is zero. It follows from the cofactor expansion of $\det(A^*)$ along the jth row that

$$\begin{aligned} 0 = \det(A^*) &= a_{i1}A^*_{j1} + a_{i2}A^*_{j2} + \cdots + a_{in}A^*_{jn} \\ &= a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn} \end{aligned}$$

Type III Row/Col Op

When **replacing** any row (or col.) of A by the sum of a scalar multiple of another row(or col.) and itself: $\beta R_i + R_j \to R_j$, $\det(E) = 1$

$$\det \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \cdots & \cdots & & \cdots \\ a_{j1} + \beta a_{i1} & a_{j2} + \beta a_{i2} & \cdots & a_{jn} + \beta a_{in} \\ \cdots & \cdots & & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

$$= \det \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \cdots & \cdots & & \cdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \cdots & \cdots & & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = 1 \cdot \det(A).$$

$\det(EA) = \det(E)\det(A)$ where $\det(E) = 1$.

Summary If E is an elementary matrix, then $\det(EA) = \det(E)\det(A)$ where

$$\det(E) = \begin{cases} -1, & \text{if } E \text{ is of type I} \\ \alpha \neq 0, & \text{if } E \text{ is of type II} \\ 1, & \text{if } E \text{ is of type III} \end{cases}$$

switch

multiply a

complicated

in column $\Rightarrow$ the same.

Summary If E is an elementary matrix, then $\det(EA) = \det(E)\det(A)$ where

$$\det(E) = \begin{cases} -1, & \text{if } E \text{ is of type I} \\ \alpha \neq 0, & \text{if } E \text{ is of type II} \\ 1, & \text{if } E \text{ is of type III} \end{cases}$$

Similar results hold for column operations:
Property If E is an elementary matrix, then $\det(AE) = \det(A)\det(E)$.

Summary If E is an elementary matrix, then $\det(EA) = \det(E)\det(A)$ where

$$\det(E) = \begin{cases} -1, & \text{if } E \text{ is of type I} \\ \alpha \neq 0, & \text{if } E \text{ is of type II} \\ 1, & \text{if } E \text{ is of type III} \end{cases}$$

Similar results hold for column operations:
Property If E is an elementary matrix, then $\det(AE) = \det(A)\det(E)$.

Proof If E is an elementary matrix, then E^T is also an elementary matrix. Then

$$\begin{aligned} \det(AE) &= \det\left((AE)^T\right) \\ &= \det\left(E^T A^T\right) \\ &= \det\left(E^T\right)\det\left(A^T\right) \\ &= \det(E)\det(A) \end{aligned}$$

Summary

The effects of row or column operations have on the value of the determinant:

1. Interchanging two rows or columns of a matrix changes the sign of the determinant.

2. Multiplying a single row or column of a matrix by a scalar has the effect of multiplying the value of the determinant by that scalar.

3. Adding a multiple of one row or column to another does not change the value of the determinant.

Example Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Find $\det(A)$ and $\det(3A)$.

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$$\det(3A) = \det\left(3\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\right) = \det\begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix} = 3\det\begin{pmatrix} 1 & 2 \\ 9 & 12 \end{pmatrix}$$

$$= 3 \cdot 3\det\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = 3^2(-2) = -18,$$

which is $3^2\det(A)$.

why? $\Rightarrow$ every row is multiplied by α.

$A : n \times n$

$\det(\alpha A) = \alpha^n \det(A)$

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$$= 3 \cdot 3\det\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = 3^2(-2) = -18,$$

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Example
$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 8 & 6 \\ 7 & 8 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & -6 \\ 0 & -6 & -12 \end{vmatrix} \quad (-4R_1 + R_2 \to R_2, -7R_1 + R_3 \to R_3)$$
$$= (-1)\begin{vmatrix} 1 & 2 & 3 \\ 0 & -6 & -12 \\ 0 & 0 & -6 \end{vmatrix} \quad (R_2 \leftrightarrow R_3)$$
$$= -36$$

Example Re-visit

Example Compute $\det(C)$ for $C = \begin{pmatrix} 0 & 0 & 3 \\ 0 & 4 & 1 \\ 2 & 3 & 1 \end{pmatrix}$.

Extra Exercises*

1. $\begin{vmatrix} 1 & 1 & 1 & 3 \\ 0 & 3 & 1 & 1 \\ 0 & 0 & 2 & 2 \\ -1 & -1 & -1 & 2 \end{vmatrix}$

2. $\begin{vmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{vmatrix}$

Extra Exercises*

1. $\begin{vmatrix} 1 & 1 & 1 & 3 \\ 0 & 3 & 1 & 1 \\ 0 & 0 & 2 & 2 \\ -1 & -1 & -1 & 2 \end{vmatrix}$

2. $\begin{vmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{vmatrix}$

Answer: 30 and -1. Simplify the problem by elementary row/col operations first: changing it into a triangular matrix.

Example

$$\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

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$$\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ x & y-x & z-x \\ x^2 & y^2-x^2 & z^2-x^2 \end{vmatrix} \quad -C_1 + C_2 \to C_2, \ -C_1 + C_3 \to C_3$$

$$= \begin{vmatrix} y-x & z-x \\ y^2-x^2 & z^2-x^2 \end{vmatrix} \quad \text{expand along the first row}$$

$$= \begin{vmatrix} y-x & z-x \\ (y-x)(y+x) & (z-x)(z+x) \end{vmatrix}$$

$$= (y-x)(z-x) \begin{vmatrix} 1 & 1 \\ y+x & z+x \end{vmatrix}$$

$$= (y-x)(z-x)(z-y)$$

More Properties of Determinant

Theorem (Re-visit) Let A be an $n \times n$ matrix,

(i) If A has a row or column consisting entirely of zeros, then $\det(A) = 0$.

(ii) If A has two identical rows or two identical columns, then $\det(A) = 0$.

Exercises: Prove the theorem.

Tips: (i) Expand along the row or column that **contains the most zeros**.
(ii) Use Type I row operation.

Theorem (iii) A determinant can be expressed as the sum of two determinants by expressing every element in any row (or column) as the sum of two terms. For example,

$$\det \begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} & a_{13}+b_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$= \det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} + \det \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

Extra Exercises*

Factorize

1. $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$
2. $\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$
3. $\begin{vmatrix} a+b & b+c & c+a \\ a^2+b^2 & b^2+c^2 & c^2+a^2 \\ a^3+b^3 & b^3+c^3 & c^3+a^3 \end{vmatrix}$

Answer: (a) $(b-a)(c-a)(c-b)$. (b) $(a+b+c)(a^2+b^2+c^2-ac-ab-bc)$. (c) $2abc(b-a)(c-a)(c-b)$

Theorem A matrix A is singular if and only if $\det(A) = 0$.

Theorem A matrix A is singular if and only if $\det(A) = 0$.

Proof Let U be the reduced row echelon form of A. Then there exists a sequence of elementary matrices E_i's, such that $U = E_k E_{k-1} \cdots E_1 A$.
It follows that

$$\begin{aligned} \det(U) = \det(E_k E_{k-1} \cdots E_1 A) &= \det(E_k)\det(E_{k-1} \cdots E_1 A) \\ &\vdots \\ &= \det(E_k)\det(E_{k-1}) \cdots \det(E_1)\det(A) \end{aligned}$$

Since $\det(E_i) \neq 0$, then $\det(A) = 0$ if and only if $\det(U) = 0$.

- If A is singular, then the last row of U must be 0, and hence $\det(U) = 0$.
- If A is nonsingular, then $U = I$ and $\det(U) = \det(I) = 1$, which implies $\det(A) \neq 0$.

From now on, we have one more equivalent condition for nonsingularity.



Theorem (Equivalent Conditions for Nonsingularity)

Let A be an $n \times n$ matrix. The following are equivalent:

(a) A is nonsingular;

(b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{0}$;

(c) A is row equivalent to I. (I is the reduced row echelon form of A. A can be written as a product of elementary matrices.)

(d) The system $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $\mathbf{b} \in \mathbf{R}^m$,

(e) $\det A \neq 0$.

Theorem If A and B are $n \times n$ matrices, then $\det(AB) = \det(A)\det(B)$.

Theorem If A and B are $n \times n$ matrices, then $\det(AB) = \det(A)\det(B)$.

Proof when B is nonsingular If B is nonsingular, B can be written as a product of elementary matrices, i.e. $B = E_k E_{k-1} \cdots E_1$ where E_i are elementary. Thus,

$$\begin{aligned}
\det(AB) &= \det(AE_k E_{k-1} \cdots E_1) \\
&= \det(A)\det(E_k)\det(E_{k-1}) \cdots \det(E_1) \\
&= \det(A)\det(E_k E_{k-1} \cdots E_1) \\
&= \det(A)\det(B)
\end{aligned}$$

Proof when B is singular Exercise

If B singular $\cdot \det(B) = 0$

then $\exists \ x \neq 0 \cdot$ s.t. $BX = 0$

$\Rightarrow (AB)X = A(BX) = 0$

$\Rightarrow AB$ singular

$\det(AB) = 0$.

$\Rightarrow \det(AB) = \det(A)\det(B)$

---

prove. $\det(A_1 \cdots A_k) = \det(A_1) \cdots \det(A_k)$

by Mathematical Induction.

Theorem Let $\left(\begin{array}{c|c} A_{k\times k} & B_{k\times l} \\ \hline C_{l\times k} & D_{l\times l} \end{array}\right)$ be a $(k+l)\times(k+l)$ block matrix. If $C_{l\times k} = O_{l\times k}$, then

$$\det\left(\begin{array}{c|c} A_{k\times k} & B_{k\times l} \\ \hline O_{l\times k} & D_{l\times l} \end{array}\right) = \det(A)\det(D).$$

Similarly, if $B_{k\times l} = O_{k\times l}$, then

$$\det\left(\begin{array}{c|c} A_{k\times k} & O_{k\times l} \\ \hline C_{l\times k} & D_{l\times l} \end{array}\right) = \det(A)\det(D).$$

Proof

$$\begin{aligned} \det\left(\begin{array}{c|c} A_{k\times k} & B_{k\times l} \\ \hline O_{l\times k} & D_{l\times l} \end{array}\right) &= \det\left(\left[\begin{array}{c|c} I_{k\times k} & O_{k\times l} \\ \hline O_{l\times k} & D_{l\times l} \end{array}\right]\left[\begin{array}{c|c} A_{k\times k} & B_{k\times l} \\ \hline O_{l\times k} & I_{l\times l} \end{array}\right]\right) \\ &= \det\left(\begin{array}{c|c} I_{k\times k} & O_{k\times l} \\ \hline O_{l\times k} & D_{l\times l} \end{array}\right)\det\left(\begin{array}{c|c} A_{k\times k} & B_{k\times l} \\ \hline O_{l\times k} & I_{l\times l} \end{array}\right) \\ &= \det(A)\det(D) \end{aligned}$$

The proof when $B_{k\times l} = O_{k\times l}$ is similar.

Extra Example*

$$\det \left[\begin{array}{cc|cc} 1 & 2 & 5 & 6 \\ 3 & 4 & 7 & 8 \\ \hline 0 & 0 & 9 & 10 \\ 0 & 0 & 11 & 12 \end{array}\right] = (1 \cdot 4 - 2 \cdot 3)(9 \cdot 12 - 11 \cdot 10) = 4$$

Extra Example*

$$\det \left[\begin{array}{cc|cc} 1 & 2 & 5 & 6 \\ 3 & 4 & 7 & 8 \\ \hline 0 & 0 & 9 & 10 \\ 0 & 0 & 11 & 12 \end{array}\right] = (1 \cdot 4 - 2 \cdot 3)(9 \cdot 12 - 11 \cdot 10) = 4$$

$\det \left[\begin{array}{c|c} A_{k\times k} & B_{k\times k} \\ \hline C_{k\times k} & D_{k\times k} \end{array}\right]$, $\det(AD - BC)$ and $\det(A)\det(D) - \det(B)\det(C)$ are not equal in general.

$$\det \left[\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \neq \det \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \det \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} - \det \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \det \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\neq \det \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right)$$

Chapter 2 Determinants

Section 2.3 Cramer's Rule

Definition (Adjoint) Let A be an $n \times n$ matrix. We define a new matrix, called the **adjoint** of A by

$$\operatorname{adj}(A) = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & & \vdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{pmatrix}^{T}.$$

Thus, to form the adjoint, we replace each entry by its cofactor and then transpose the resulting matrix.

Theorem Let A be a nonsingular $n \times n$ matrix, that is, $\det(A) \neq 0$. Then

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

Proof By a lemma in Sc2.2,

$$A(\operatorname{adj}(A)) = \det(A) I_n.$$

If A is nonsingular, $\det(A) \neq 0$, and we may write

$$A\left(\frac{1}{\det(A)} \operatorname{adj}(A)\right) = I_n.$$

Thus

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A), \quad \text{when } \det(A) \neq 0.$$

$$\begin{bmatrix} \det(A) & 0 \\ \vdots & \vdots \\ \cdots & \cdots \end{bmatrix}$$

$$= \begin{bmatrix} \det(A) & \cdots & 0 \\ 0 & \ddots & \\ 0 & & \det(A) \end{bmatrix} = \det(A) \cdot I_n$$

$$\Rightarrow A^{-1} = \frac{\text{adj}(A)}{\det(A)}$$

$$\det(A) = \begin{cases} a_{11} & \text{if } n = 1 \\ a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1n}A_{1n} & \text{if } n > 1 \end{cases},$$

linear equation $Ax = b$ → 已知. 已知. (coefficient)

$$A^{-1} \cdot Ax = Ix = A^{-1} \cdot b$$

$$\star\ x = A^{-1} \cdot b$$

Example Find the inverse of $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $\det(A) \neq 0$.

Solution

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$\begin{matrix} a & b \\ c & d \end{matrix}$

$\text{adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$A_{11} \begin{bmatrix} d & -c \\ -b & a \end{bmatrix} A_{12} \Rightarrow$ Transpose

$A_{21} \quad A_{22}$ $\Rightarrow \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Example re-visit

Let $A = \begin{bmatrix} 0 & 2 & -8 \\ 1 & -2 & 1 \\ -4 & 5 & 9 \end{bmatrix}$. Find A^{-1} using adjoint of A. Hence solve

$$\begin{bmatrix} 0 & 2 & -8 \\ 1 & -2 & 1 \\ -4 & 5 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{pmatrix} 8 \\ 0 \\ 9 \end{pmatrix}$$

Answer:

$$A^{-1} = \begin{bmatrix} 23/2 & 29 & 7 \\ 13/2 & 16 & 4 \\ 3/2 & 4 & 1 \end{bmatrix}, \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1}b = \begin{bmatrix} 23/2 & 29 & 7 \\ 13/2 & 16 & 4 \\ 3/2 & 4 & 1 \end{bmatrix} \begin{bmatrix} 8 \\ 0 \\ 9 \end{bmatrix} = \begin{bmatrix} 29 \\ 16 \\ 3 \end{bmatrix}$$

square matrix, nonsingular

Theorem (Cramer's Rule) Let A be an nonsingular $n \times n$ matrix, and let $\mathbf{b} \in \mathbf{R}^n$. Let A_i be the matrix obtained by replacing the i-th column of A by $\mathbf{b}$. If $\mathbf{x}$ is the unique solution of $A\mathbf{x} = \mathbf{b}$, then

$$x_i = \frac{\det(A_i)}{\det(A)}, \quad \text{for } i = 1, \cdots, n.$$

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1i} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2i} & \cdots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & a_{ni} & \cdots & a_{nn} \end{pmatrix} \xrightarrow[\text{by } \mathbf{b}]{\text{Replace } i\text{-th col.}} A_i = \begin{pmatrix} a_{11} & \cdots & b_1 & \cdots & a_{1n} \\ a_{21} & \cdots & b_2 & \cdots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & b_n & \cdots & a_{nn} \end{pmatrix}$$

Proof Since

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{\det(A)}(\operatorname{adj}(A))\mathbf{b}.$$

it follows that

$$x_i = \frac{b_1A_{1i} + b_2A_{2i} + \cdots + b_nA_{ni}}{\det(A)} = \frac{\det(A_i)}{\det(A)}.$$

$$(\operatorname{adj})A\mathbf{b} = \begin{pmatrix} b_1A_{11} & b_2A_{21} & \cdots & b_nA_{n1} \\ b_1A_{12} & & & \\ \vdots & & & \\ b_1A_{1n} & b_2A_{2n} & \cdots & b_nA_{nn} \end{pmatrix}$$

Example Consider

$$\begin{pmatrix} 0 & 2 & -8 \\ 1 & -2 & 1 \\ -4 & 5 & 9 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ 0 \\ -9 \end{pmatrix}$$

$n \times n$. nonsingular

Solution $\det(A) = -2$.

change column

$$|A_1| = \begin{vmatrix} 8 & 2 & -8 \\ 0 & -2 & 1 \\ -9 & 5 & 9 \end{vmatrix} = -58, \quad |A_2| = \begin{vmatrix} 0 & 8 & -8 \\ 1 & 0 & 1 \\ -4 & -9 & 9 \end{vmatrix} = -32, \quad |A_3| = \begin{vmatrix} 0 & 2 & 8 \\ 1 & -2 & 0 \\ -4 & 5 & -9 \end{vmatrix} = -6$$

We obtain $x = \dfrac{|A_1|}{|A|} = 29, \ y = \dfrac{|A_2|}{|A|} = 16, \ z = \dfrac{|A_3|}{|A|} = 3.$

Exercise Use Cramer's rule to solve

$$\begin{array}{llll} x_1 & +2x_2 & +x_3 & = 5 \\ 2x_1 & +2x_2 & +x_3 & = 6 \\ x_1 & +2x_2 & +3x_3 & = 9 \end{array}$$

Answers: 1,1,2.

$$-4 \leftarrow \begin{bmatrix} 1 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ 9 \end{bmatrix}$$

$$x_1 = \frac{1 \times 4 - 2 \times 9 - 6}{-4} = 1$$

$x_2 = \frac{48 - 9 - 5\times(6-1) + 18 - 6}{-4} = 1$

$x_3 = \frac{18 - 12 - 2\times(18-6) + 5\times 2}{-4}$

$= \frac{-8}{-4} = 2$

Chapter 2 Determinants

Section 2.3 Cramer's Rule

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Answers: 1,1,2.

Chp 2 Review

Linear Equations:

=> matrices => augmented matrix

upper triangular form

Row Reduction → Row Echelon Form (ref) → back substitution

$m \cdot n$ (2D(3D)) => 是否有解. ✗ Ⓥ ／ Ⓧ

↳ 2-3

∀

partioned matrix